RIT | Kosovo

PROGRAMMING WITH AI

TRAINING CURRICULUM

Total Duration	30 Hours
Level	Beginner to Intermediate
Training Format	Lectures, demonstrations, exercises, group activities, and a final project
Language	English

Professional Development Program

1. Course Description

The Programming with AI training program introduces participants to the fundamental concepts of programming with Python and the practical use of modern artificial intelligence tools. The curriculum combines essential programming principles with large language models and AI-powered platforms. Participants learn how to write, analyze, test, and improve code while using artificial intelligence as a supporting tool throughout the software development process.

2. Course Objectives

- Introduce participants to the fundamental principles of programming
- Develop practical Python programming skills
- Build logical thinking and problem-solving abilities
- Introduce commonly used artificial intelligence platforms
- Explain how large language models process and generate information
- Develop effective prompt-writing skills
- Demonstrate how AI can support software development
- Introduce chatbots, AI tools, automations, and AI agents
- Promote responsible, secure, and ethical use of artificial intelligence
- Support participants in developing a practical programming project

3. Learning Outcomes

- Understand the basic principles of programming
- Write and execute Python programs
- Use variables and different data types
- Add and understand comments in source code
- Accept and process user input
- Use conditional statements for decision-making
- Implement for and while loops
- Work with lists, tuples, dictionaries, and sets
- Create and use functions
- Understand the basic principles of object-oriented programming
- Create simple classes and objects
- Identify the main differences between popular AI platforms
- Write clear and effective prompts
- Understand tokens, context windows, and model limitations
- Use AI tools to generate, explain, debug, and improve code
- Understand the differences between chatbots, AI tools, automations, and AI agents
- Recognize privacy, security, and ethical risks when using AI
- Develop a simple Python application supported by artificial intelligence

4. Course Syllabus

Module 1: Introduction to Programming and Python

- Definition of programming
- Role of programming in software development
- Introduction to Python
- Main characteristics of Python
- Common applications of Python
- Installing Python
- Installing and configuring Visual Studio Code
- Creating and saving Python files
- Running Python programs
- Understanding basic syntax
- Using the Python terminal
- Identifying and correcting basic syntax errors

Module 2: Variables and Comments

- Definition of a variable
- Creating and assigning variables
- Variable naming conventions
- Valid and invalid variable names
- Updating variable values
- Assigning multiple variables
- Understanding constants as a programming convention
- Writing single-line comments
- Writing multi-line explanations
- Using comments to improve code readability
- Importance of clear and meaningful variable names

Module 3: Input and Output

- Displaying information with the print() function
- Printing text, numbers, and variables
- Combining text and variables
- Using formatted strings
- Understanding the input() function
- Receiving information from the user
- Storing user input in variables
- Converting user input into appropriate data types
- Creating simple interactive programs
- Validating basic user input

Module 4: Python Data Types

- Understanding data types
- Strings
- Integers
- Floating-point numbers
- Boolean values
- Checking data types with type()
- Converting between data types
- String concatenation
- Basic string methods
- Basic arithmetic operations
- Comparison operators
- Logical operators
- Understanding mutable and immutable values

Module 5: Conditional Statements

- Introduction to decision-making in programming
- Using the if statement
- Using the else statement
- Using the elif statement
- Writing multiple conditions
- Using comparison operators in conditions
- Using logical operators
- Creating nested conditional statements
- Validating user input with conditions
- Developing simple decision-based programs
- Understanding indentation in Python
- Avoiding common conditional statement errors

Module 6: Loops

- Purpose of loops in programming
- Understanding repeated execution
- Using the while loop
- Defining loop conditions
- Updating loop variables
- Preventing infinite loops
- Using the for loop
- Iterating through strings and collections
- Using the range() function
- Controlling loops with break
- Skipping iterations with continue
- Using nested loops
- Selecting the appropriate loop for a task
- Solving repetition-based programming exercises

Module 7: Data Structures

- Lists: creating lists, indexes, updating values, adding and removing elements, sorting, list length, iteration, and common methods
- Tuples: creating tuples, accessing elements, immutability, tuple packing and unpacking, and appropriate use cases
- Dictionaries: keys and values, access, addition, updates, removal, iteration, methods, and structured records
- Sets: unique values, adding and removing elements, duplicate removal, union, intersection, difference, and comparison with other structures

Module 8: Functions

- Purpose of functions
- Defining functions with def
- Calling functions
- Creating reusable blocks of code
- Using function parameters
- Using multiple parameters
- Understanding arguments
- Returning values with return
- Difference between printing and returning values
- Using default parameter values
- Understanding local and global variables
- Creating functions for calculations
- Creating functions for data validation
- Organizing programs with multiple functions
- Writing clear function names and documentation

Module 9: Introduction to Object-Oriented Programming

- Introduction to object-oriented programming
- Understanding classes

- Understanding objects
- Creating a class
- Creating objects from a class
- Defining attributes
- Defining methods
- Using the __init__() method
- Understanding the self parameter
- Passing values when creating objects
- Updating object attributes
- Calling object methods
- Creating simple real-world class examples
- Comparing procedural and object-oriented programming
- Organizing code with classes

Module 10: Introduction to Artificial Intelligence

- Definition of artificial intelligence
- Brief development of artificial intelligence
- Difference between traditional software and AI systems
- Introduction to machine learning
- Introduction to generative AI
- Understanding large language models
- Common applications of artificial intelligence
- AI in education
- AI in software development
- AI in business and communication
- Benefits and limitations of AI systems
- Human supervision in AI-assisted work

Module 11: Popular AI Platforms

- ChatGPT: common use cases, text generation, summarization, code generation, file work, strengths, and limitations
- Google Gemini: Google ecosystem integration, multimodal capabilities, research, productivity, strengths, and limitations
- Claude: long-document processing, code analysis, structured writing, reasoning tasks, strengths, and limitations
- DeepSeek: programming use cases, code generation, technical problem-solving, strengths, and limitations
- Kimi: long-context processing, document analysis, research tasks, information organization, strengths, and limitations
- Platform comparison: usability, code generation, document processing, response quality, and task-based selection

Module 12: Prompt Engineering

- Definition of a prompt
- Importance of clear instructions
- Writing specific prompts
- Providing relevant context
- Defining the expected output
- Assigning a role to the AI model
- Adding examples to prompts
- Using constraints and requirements
- Requesting structured output
- Improving weak prompts
- Developing prompts step by step
- Asking follow-up questions
- Evaluating AI-generated responses
- Avoiding vague and incomplete instructions
- Creating prompts for programming tasks

Module 13: Tokens and Context

- Definition of a token

- How text is divided into tokens
- Relationship between tokens and words
- Input tokens
- Output tokens
- Understanding the context window
- Why long conversations may lose earlier information
- Managing large prompts
- Reducing unnecessary information
- Structuring context effectively
- Understanding output limitations
- Basic awareness of API usage and token-based costs
- Writing efficient prompts

Module 14: Chatbots, AI Tools, Automations, and Agents

- Chatbots: definition, message-based interaction, rule-based and AI-powered chatbots, common use cases, and limitations
- AI tools: task-specific applications, writing, summarization, image generation, programming assistants, research tools, and human-in-the-loop workflows
- Automations: repetitive workflows, triggers, schedules, data transfer, and differences between automation and AI
- AI agents: goals, planning, decision-making, tool use, memory, multi-step task completion, supervision, and comparison with chatbots

Module 15: Using AI for Programming

- Generating simple Python code
- Explaining existing code
- Translating requirements into code
- Identifying syntax errors
- Debugging logical errors
- Improving code readability
- Refactoring code
- Generating comments and documentation
- Creating test cases
- Asking AI to explain errors
- Reviewing AI-generated code
- Testing code before using it
- Recognizing incorrect or unsafe suggestions
- Combining human reasoning with AI assistance

Module 16: APIs and AI Integration Fundamentals

- Definition and purpose of an API
- Communication between applications
- Request and response concepts
- Basic understanding of HTTP
- Introduction to JSON
- API keys and environment variables
- Protecting credentials
- Basic structure of an AI API request
- Sending user input to an AI service
- Receiving and displaying an AI-generated response
- Understanding basic error responses
- Difference between an AI website and an AI API
- Overview of connecting Python applications with AI services

Module 17: Responsible and Ethical Use of AI

- Responsible use of artificial intelligence
- Accuracy and reliability of AI-generated information
- AI hallucinations
- Verification of generated content

- Bias in AI systems
- Copyright and intellectual property
- Academic integrity
- Proper acknowledgment of AI assistance
- Privacy and personal data
- Personally identifiable information
- Confidential business information
- Password and API key protection
- Risks of uploading sensitive documents
- Human responsibility for final decisions
- Ethical use of generated code and content

Module 18: Practical Exercises and Problem-Solving

- Creating programs with variables and user input
- Building simple calculators
- Developing conditional decision programs
- Creating programs with for loops
- Creating programs with while loops
- Managing collections with lists
- Creating structured information with dictionaries
- Removing duplicate data with sets
- Writing reusable functions
- Creating simple classes and objects
- Generating code with AI assistance
- Reviewing and correcting AI-generated code
- Comparing answers from different AI platforms
- Improving prompts based on response quality
- Solving small programming challenges

Module 19: Final Project

- Development of a simple Python application that combines the programming concepts covered during the training with the responsible use of artificial intelligence
- Project components may include user input, variables, data types, conditionals, loops, data structures, functions, classes, validation, error handling, AI-assisted functionality, comments, and documentation
- Possible project ideas: AI-assisted study assistant, FAQ chatbot, personal task organizer, student information system, quiz application, recommendation application, text summarization assistant, customer support prototype, code explanation assistant, or interactive educational chatbot

5. Teaching Methodology

- Instructor-led presentations
- Live coding demonstrations
- Guided practical exercises
- Individual programming tasks
- Group discussions
- Problem-solving activities
- AI platform demonstrations
- Prompt-writing exercises
- Code review activities
- Project-based learning
- Continuous feedback and support

6. Assessment Methods

- Active participation
- Completion of practical exercises
- Short programming assignments
- Prompt-writing activities

- Code analysis tasks
- Debugging exercises
- Practical demonstrations
- Final project development
- Final project presentation

7. Required Tools and Platforms

- Computer with internet access
- Python
- Visual Studio Code
- Web browser
- Git
- GitHub account
- ChatGPT
- Google Gemini
- Claude
- DeepSeek
- Kimi
- An AI API platform when required for demonstrations

8. Final Competencies

- Basic Python programming competence
- Logical and structured problem-solving
- Ability to create small Python applications
- Ability to read and understand Python code
- Ability to use data structures appropriately
- Ability to create functions and basic classes
- Understanding of major AI concepts
- Effective prompt-writing skills
- Ability to compare and select AI tools
- Ability to use AI for programming support
- Awareness of security, privacy, and ethical responsibilities
- Ability to present and explain a completed programming project

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